

How Are PROGRESS-Plus Subpopulations Represented in Systematic Reviews Evaluating Illicit Drug Policy Interventions? A Review of Reviews

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Abstract

Background: Current evidence suggests that certain subpopulations bear a disproportional burden of illicit drug use and related comorbidities. With drug use being a risk factor for many chronic and acute diseases, addressing this health determinant could reduce health disparities prior to their clinical manifestation. This umbrella review assesses the representation of at-risk subpopulations in systematic reviews evaluating illicit drug interventions. It further aims to identify intervention characteristics associated with effective policy design for individuals experiencing structural disadvantage factors.

Methods: Three databases were searched for systematic reviews and meta-analyses of the effectiveness of drug use policy interventions among subpopulations experiencing one or more relevant PROGRESS-Plus risk-factors. The data was double extracted, and quality was assessed using AMSTAR 2 quality assessment tool. Results were synthesized narratively.

Results: Fifty-three studies were included in the review, of which 28 evaluated drug interventions for individuals experiencing one PROGRESS-Plus risk factor (i.e. non-white race) and 25 evaluated intervention effectiveness for individuals experiencing multiple co-occurring risk factors (i.e. non-white race and rural residence). The most represented domain was age (children/adolescents), place of residence (rural/inner city/vulnerably-housed) and race, ethnicity, or culture (racial/indigenous minorities). We found some evidence of effectiveness of targeted policies, especially for ethnic minorities, who may not benefit proportionately from universal interventions alone.

Conclusions: Our findings point equity-concerned decision makers towards targeted universalism when designing interventions, and to consider supplementing universal interventions with program adaptations for minority sub-populations. We identified the following PROGRESS-Plus subgroups as under-represented in existing knowledge syntheses: religious minorities, individuals with disabilities, low educational attainment, socially isolated and unemployed individuals. Further research is needed to fill this subgroup-specific gap in knowledge syntheses to ensure that decision-makers interested consider how their interventions affect all relevant stakeholders.

Keywords: *Health equity, drug use, policy intervention, public health, targeted universalism, umbrella review*

1 Introduction

The argument that public policy design affects the distribution of population health and health inequities has long been recognized and supported by a wide base of research evidence (Arblaster et al. 1996, Naik et al. 2019). Over the last two decades, research production and publication on the effects of

policy interventions on health-related outcomes grew rapidly. At the same time, social sciences witnessed a surge in the use of systematic review and meta-analysis methodology. This has been an important step towards meeting the goals set by the World Health Organization (WHO) Commission on Social Determinants of Health (CSDH) and actively addressing health inequities, because availability of rigorous, extensive, and high-quality evaluations of health policy effectiveness, such as those offered by well-conducted systematic review,

are essential to address health inequities. However, it is crucial to move beyond evaluating interventions based on their average effects on health and towards addressing intervention impact with respect to different population subgroups. This is because not all relevant stakeholders may benefit equally from the interventions available and using mean policy effects may mask the potential of these policies to perpetuate or even increase existing health inequities.

This potential problem with policy evaluations is especially pressing in the broader context of illicit drug use and substance use disorders (SUD), which, as recent research suggests, are unequally distributed across socio-economic groups, genders, or ethnicities, and, at the same time, prone to differential intervention effectiveness across these population subgroups (Patrick et al. 2012, NIDA 2020b, McCabe et al. 2007, Cao, Burton, and Liu 2018). With illicit drug use being a major adjustable risk factor for many chronic and acute diseases, addressing how well different drug policies work for subpopulations could help us reduce social inequalities in health outcomes related to drug use before the disease onset becomes a critical issue both in itself and as a health equity concern (National Center for Health Statistics (U.S.) 2012, Pampel, Krueger, and Denney 2010, Petrovic et al. 2018). The importance of not only preventing illicit drug use overall, but also understanding whether and how specific policies work for different disadvantaged minority groups, is especially pressing considering that preventable disparities in health outcomes constitute a serious ethical, social, and political concern. To effectively address this complex issue, it is crucial to provide decision-makers with well-conducted systematic reviews that evaluate illicit drug policy effectiveness with respect to different subpopulations deemed at-risk. Such knowledge syntheses represent the best available evidence to equity-concerned decision makers. Their lack thereof, on the other hand, indicates a strong possibility of an insufficient primary research output with respect to the excluded subpopulations at elevated risk of drug use and/or its comorbidities.

We were unable to identify a comprehensive overview of existing systematic reviews on the effectiveness of illicit drug interventions among subpopulations at risk. Given the importance of systematic reviews in informing evidence-based decision makers and the existing evidence pointing to a non-uniform intervention effectiveness among different population subgroups, it is important to understand whether these subpopulations at-risk are specifically considered in relevant systematic reviews.

This umbrella review aims to provide a comprehensive synthesis of the existing systematic reviews of illicit drug intervention effectiveness among different subpopulations at elevated risk of harmful illicit drug use and differential intervention effectiveness. By embedding the PROGRESS-Plus classification framework for common disadvantage factors within the umbrella review, we have adopted a novel and adaptable methodological way of assessing whether relevant subpopulations at-risk are included in impactful systematic reviews, which could be used within any policy evaluation field.

Research Objective

The main objective of the study is to synthesize and classify the existing knowledge of the effectiveness of illicit drug policy interventions among relevant PROGRESS-Plus population subgroups from high-income countries OECD and non-OECD EU-28 countries.

2 Methods

This paper used umbrella review methodology to synthesize evidence from peer-reviewed systematic reviews. The protocol for this review was developed prior to the search and registered with PROSPERO (CRD42021251045) in January 2021. As per the standard practice in evidence-based syntheses, inclusion criteria were defined a priori in terms of the ‘Population, Intervention, Comparison, Outcome and Study Design’ (PICOS) format (Higgins & Thomas, 2021). For full inclusion and exclusion criteria, see Table 1 below.

Table 4: Systematic reviews of interventions for individual PROGRESS-Plus factors

PROGRESS-Plus Domain	Review	Setting	Population	Outcome of Interest	Policy Type	Intervention Details	Policy effect	N° of relevant studies
Place of Residence	(O'Mara-Eves et al., 2015)	OECD HIC	Inter-city or rural	Drug Use Frequency	Universal	Public-health interventions such as community-setting interventions	0	18 (131)
	(Puska et al., 2020)		Homeless or vulnerably housed population	Drug use reduction	Targeted	Case Management Interventions (Intensive vs Standard)	+/-	22 (56)
	(de Vet et al., 2013)	USA, CAN			Targeted	Emergency department interventions, including case management interventions for drug use reduction (intensive vs standard case management)	*	4 (21)
Race/ Ethnicity/ Culture	(McLean et al., 2020)	USA	Indigenous population	Drug use prevention	Targeted	Culturally grounded, deep-structure prevention interventions	+	5 (14)
	(Gorin et al., 2012)		Non-white black and Hispanic population	Interventions drug use	Targeted	Multi-level targeted education and training interventions in schools through external consultant delivery	+/*	3 (26)
	(Guerrero et al., 2013)			Drug use frequency	Targeted	Culturally appropriate and language-adapted motivational enhancements and integrated treatments for substance abuse disorders at community level	*	5 (100)
	(O'Mara-Eves et al., 2015)	OECD HIC			Universal	Large-scale public-health interventions such as community-setting interventions	+	18 (131)
	(Hefner et al., 2007)	USA	Hispanic, Latino population	Drug use and HIV risk	Targeted	HIV behavioral prevention interventions, personal skills referrals to social services, job placements in Hispanic communities, adapted to Hispanic culture and Spanish	+	7 (20)
Socio-economic Position	(O'Mara-Eves et al., 2015)	OECD HIC	Low socio-economic status population	Drug Use	Targeted	Public-health interventions such as community-setting interventions, tailored mass media programs	+	36 (131)
Plus 1: Age	(Hale et al., 2014)	OECD HIC	Youth population	Drug use frequency	Universal	School-based interventions and family-school partnerships, web-based interventions	+/*	31 (55)
	(Mewton et al., 2018)	OECD HIC		Drug use frequency	Targeted	Targeted school and family-based interventions for youth	+	20 (52)
	(Bosar et al., 2020)	USA		Drug use frequency	Universal	Media campaigns for prevention of illicit drug use, anti-illicit drug public service announcements	*	2 (76)
	(Smedstad et al., 2017)	OECD HIC		Cannabis use	Targeted	School community-based interventions for youth and their parents	+	7 (60)
						Brief, computerized interventions for early stages of substance use delivered for a short duration of time, such as assessment, feedback, and decision-making modules	0	7 (60)

(Excerpt) Table 1: PICOS inclusion and exclusion criteria; Table 2: description of adapted PROGRESS-Plus disadvantage domains

To capture the effectiveness of policies on drug use among different subpopulations, we have chosen to use an adapted PROGRESS-Plus equity framework to classify our populations of interest into minority subgroups based on their membership in at least 1 PROGRESS-Plus domain identified as a disadvantage factor for illicit drug use. This framework allowed us to apply an equity lens to interventions and to identify characteristics that stratify health opportunities or outcomes (O'Neill et al., 2014). The adaptations to the framework were made by the study authors to reflect the relevance of each PROGRESS factor to the study of illicit drug use. Table 2 presents a list of the domains included in the framework with examples for each domain (see above). A more comprehensive literature review relating each domain to subpopulation-specific illicit drug use research is included in Appendix A.

Search strategy

A literature search strategy for existing systematic reviews was developed by the first author and peer-reviewed by a liaison librarian, Genevieve Gore. Three databases were searched (with host sites in parentheses): MEDLINE (Ovid), EMBASE (Ovid) and EPISTEMONIKOS. The searches were conducted

without time frame restrictions and selected references were imported into ENDNOTE X9. Key words searched were adapted to each database host site. A grey literature search was conducted by citation follow-up from included studies as well as through a Google Scholar search (see Appendix B for full search strategy including detailed search terms). The screening process at title/abstract level was double extracted by two independent reviewers. Conflicts in inclusion/exclusion decision were resolved by discussion. There was a high agreement in both parts of the screening (abstract: 85.4% and full text: 87.9%). A precautionary approach was used to include studies unless exclusion was clearly justified by the exclusion criteria listed in Table 1. A kappa statistic was not calculated, for in cases of a relatively low inclusion rate compared to the number of papers found, the score is not appropriate (Naik et al., 2019). Full text review, data extraction and quality appraisal were conducted independently by two authors, MB and YD. Figure 1 below shows a PRISMA flow chart of the distinct phases of this umbrella review.

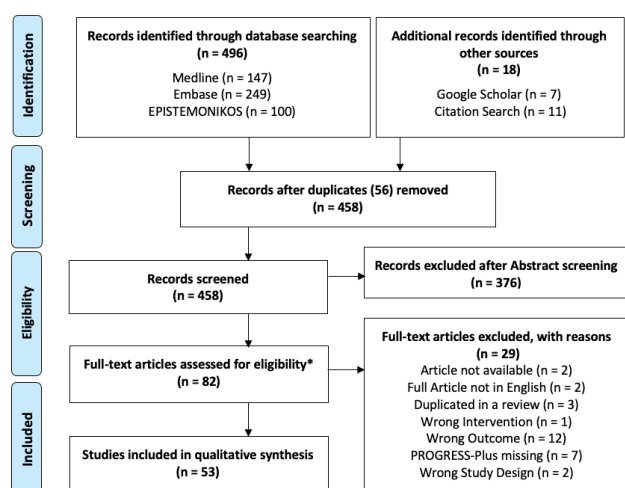


Figure 1. PRISMA flow chart.

Data synthesis and quality appraisal

Data extraction was conducted by filling out an adapted version of the Data Collection Form template for randomized controlled trials (RCT) and non-RCT from the Cochrane Collaboration (Cochrane Collaboration, 2020). Where not all findings of the included reviews were relevant to the scope of our umbrella review, only relevant findings were extracted. Given the heterogeneous methodologies of included reviews, the summary effect estimates reported in our review were handled accordingly:

- *For narrative reviews:* ‘positive’ or ‘negative’ intervention effect was based on whether the authors of the systematic review reported that the majority of individual studies included in the review reached a similar conclusion about intervention effectiveness on illicit drug use or related comorbidities. We reported an ‘inconclusive’ summary effect estimate if the individual studies in the review presented findings that were imprecise, or insignificant. A ‘no’ effect size was reported when the majority of included studies

reported no effect of the intervention studied on relevant health outcomes and a ‘mixed’ effect was reported when individual studies in the review came to conflicting conclusions about the effectiveness of the intervention studied.

- *For meta-analyses:* ‘positive’ or ‘negative’ intervention effect was based on whether the meta-analysis reported a statistically significant positive or negative summary result from a statistical synthesis of individual studies. We reported an ‘inconclusive’ summary effect estimate if the statistical synthesis did not reveal a statistically significant trend in intervention effectiveness across included studies (i.e., the overall estimate was imprecise or insignificant). ‘No’ effect size was reported when the meta-analysis reported a null overall estimate for intervention effectiveness and a ‘mixed’ effect was reported (when applicable), where individual studies within a review reported statistically significant and yet contradictory conclusions.

As part of the data extraction, each review included was assessed against all 16 components of AMSTAR 2, a critical appraisal tool for systematic reviews of randomized and non-randomized studies and meta-analyses (Shea et al., 2017). The components included questions related to the quality of the search strategy, research conduct, heterogeneity, conflicts of interest and risk of bias assessment, such as selection bias or publication bias. We have chosen to judge studies as either of high, moderate or low risk of bias based on the percentage of ‘applicable’ AMSTAR 2 components answered as ‘Yes’ (see Table 3 below for details). The condition was phrased as ‘appropriate’ as opposed to ‘all’ AMSTAR 2 components, because some questions specifically target bias assessment associated with the statistical methodology of meta-analyses, which were not applicable to some narrative systematic reviews and umbrella reviews included. Scoping reviews were assessed for quality in the same way as systematic reviews, using AMSTAR 2. However, in the case of scoping reviews, questions related to the quality of reporting as well as risk of bias were not considered, because they are not necessary components of a scoping review methodology. While all 16 components were relevant to meta-analyses, only 12 were used to assess narrative systematic reviews and 10 for scoping reviews.

Table 3. AMSTAR 2 quality assessment

AMSTAR 2 rating	% of appropriate questions answered ‘Yes’
Low risk of bias	>80%
Moderate risk of bias	80–50%
High risk of bias	<50%

The findings were presented both in a tabular form for each PROGRESS-Plus disadvantage factor and summarized through a narrative synthesis. A quantitative data synthesis, such as a meta-analysis was not conducted as part of this review, due to the heterogeneity of included populations and interventions.

3 Results

Description of included reviews

We identified a total of 53 eligible reviews of illicit drug interventions effectiveness with health effect evaluation for individuals experiencing at least one relevant PROGRESS-Plus disadvantage factor (see Figure 2, Tables 4 and 5).

Twenty-seven of these studies evaluated the interventions for individuals experiencing one single PROGRESS-Plus disadvantage factor. Of these, the most represented PROGRESS-Plus subpopulations in reviews were youth population (6/27), racial or ethnic minority population (5/27), substance use disorders (5/27) and disadvantaged place of residence (4/27). Effectiveness of drug interventions for individuals experiencing multiple co-occurring PROGRESS-Plus factors was examined in 26 reviews. Most of these reviews (19/26) concentrated on age, namely children, adolescents, or specific vulnerable adults such as women of childbearing age, experiencing at least one additional PROGRESS-Plus disadvantage factor, such as vulnerable housing, ethnic or sexual minority status.

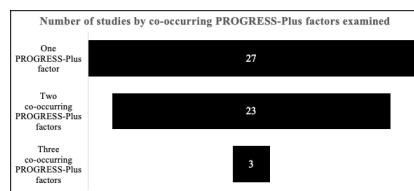


Figure 2. Number of papers by inclusion of PROGRESS-Plus factors.

Figure 3 below shows the number of reviews that explicitly evaluated illicit drug policies for each PROGRESS-Plus disadvantage subgroup. Reviews that evaluated policies for multiple co-occurring PROGRESS-Plus factors were enumerated in all respective categories and therefore, the sum of all reviews on Figure 3 is higher than the 53 included reviews. None of the included reviews looked at the effectiveness of illicit drug policy interventions for individuals with a vulnerable occupational status, low social capital, and religious minorities. Interventions for individuals with low educational attainment and socio-economic status were evaluated in only one respective review. Age, and specifically youth population, such as children, adolescents, and emerging adults, was the most represented population subgroup for which illicit drug interventions were evaluated (24/53), followed by racial, ethnic, and cultural minorities (14/53), individuals with existing substance use disorders (12/53) and individuals living in a disadvantaged place of residence (11/53).

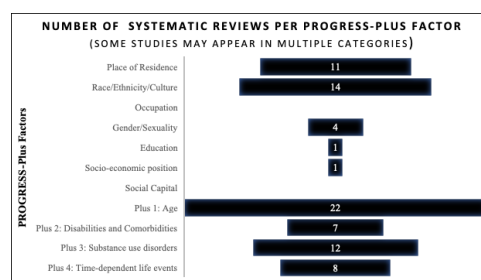


Figure 3. Number of systematic reviews per PROGRESS-Plus domain.

Despite no time restrictions, all the reviews included in this paper were published after 2005 (see Figure 4 below). Most of the included reviews were published within the last 10 years, pointing to an increasing awareness of the value of systematic reviews, advances in conducting them and potentially a growing interest in subgroup policy evaluation analysis after the publication of The Marmot Review in 2010. Most of the reviews were set exclusively in the United States (25/53), either by design or because only US primary studies were identified in the reviews. Of the additional 28 reviews, 23 explicitly focused on OECD high-income countries and 5 were without a specific country development setting, but only included high-income OECD countries. Within these reviews, most primary studies were conducted in the United States.

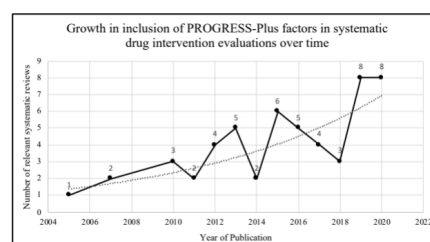


Figure 4. Growth in inclusion of PROGRESS-Plus factors in systematic drug intervention evaluations over time.

Of the included studies, 20 reviews were assessed as being at low risk of bias, 19 as being at moderate and 14 at high risk of bias using the AMSTAR 2 quality assessment tool for systematic reviews and meta-analyses. Approximately 58% of the included reviews used exclusively a systematic review methodology (31/53), 15% provided both a systematic review and a meta-analysis (8/53), and an additional 12% used a meta-analysis methodology only (7/53). We also included 3 well-conducted scoping reviews and 4 umbrella reviews that explicitly reported relevant outcomes (7/53). Most reviews included primary studies that compared the effectiveness of their intervention of interest either against no intervention, or standard treatment, using observational study design or randomized control trial design.

Excluded reviews

We have excluded 443 reviews that did not meet our inclusion criteria (see Figure 1 for a PRISMA flow chart). Most reviews (376) were excluded during the title/abstract screening process. Of the 82 reviews included in the full-text review, 29 were excluded (for exclusion reasons for these reviews, see Appendix C). Most of these reviews were rejected either because the reviews did not include a separate and quantifiable illicit drug use outcome (12), or because the outcome was not evaluated specifically for one of the PROGRESS-Plus disadvantage domains (7).

Effects of illicit drug interventions on PROGRESS-Plus subpopulations

The full data-extraction tables summarizing every included review by setting, population, outcome, policy type, intervention details, direction of effect, and number of relevant studies – for reviews addressing a single PROGRESS-Plus factor (Table 4) and for reviews addressing multiple co-occurring PROGRESS-Plus factors (Table 5) – are reproduced on the following pages.

Table 4: Systematic reviews of interventions for individual PROGRESS-Plus factors

PROGRESS-Plus Domain	Review	Setting	Population	Outcome of Interest	Policy Type	Intervention Details	Policy effect	N° of relevant studies†
Place of Residence	(O'Mara-Eves et al., 2015)	OECD HIC	Inner-city or rural	Drug Use Frequency	Universal	Public-health interventions such as community-setting interventions	0	18 (131)
	(Ponka et al., 2020)		Homeless or vulnerably housed population	Drug use reduction	Targeted	Case Management Interventions (Intensive vs Standard)	+/-	22 (56)
	(de Vet et al., 2013)						*	4 (21)
	(Formosa et al., 2021)	USA, CAN			Targeted	Emergency department interventions, including case management interventions for drug use reduction (intensive vs standard case management)	*	4 (13)
Race/Ethnicity/Culture	(McLean et al., 2020)	USA	Indigenous population	Drug use prevention	Targeted	Culturally grounded, deep-structure prevention interventions	+	5 (14)
	(Gorin et al., 2012)		Non-white black and Hispanic population	Intravenous drug use	Targeted	Multi-level targeted education and training interventions in schools through external consultant delivery	+/*	3 (26)
	(Guerrero et al., 2013)			Drug use frequency	Targeted	Culturally appropriate and language-adapted motivational enhancements and integrated treatments for substance abuse disorders at community level	*	5(100)
	(O'Mara-Eves et al., 2015)	OECD HIC			Universal	Large-scale public-health interventions such as community-setting interventions	+	18 (131)
	(Herbst et al., 2007)	USA	Hispanic, Latino population	Drug use and HIV risk	Universal	HIV behavioral prevention interventions, personal skills	+	7 (20)
					Targeted	Referrals to social services, job placements in Hispanic communities, adapted to Hispanic culture and Spanish	+	
Socio-economic Position	(O'Mara-Eves et al., 2015)	OECD HIC	Low socio-economic status population	Drug Use	Targeted	Public-health interventions such as community-setting interventions, tailored mass media programs	+	36 (131)
					Universal	Universal mass media or health care setting interventions	*	
Plus 1: Age	(Hale et al., 2014)	OECD HIC	Youth population	Drug use frequency	Universal	School-based interventions and family-school partnerships, web-based interventions	+/*	31 (55)
					Targeted	Targeted school and family-based interventions for youth	+	
	(Mewton et al., 2018)	OECD HIC		Drug use frequency	Universal	Media campaigns for prevention of illicit drug use, anti-illicit drug public service announcements	*	20 (52)
	(Bonar et al., 2020)	USA		Drug use frequency	Universal	School community-based interventions for youth and their parents	+	2 (76)
	(Smedslund et al., 2017)	OECD HIC		Cannabis use	Targeted	Brief, computerized interventions for early stages of substance use delivered for a short duration of time, such as assessment, feedback, and decision-making modules	0	7 (60)

PROGRESS-Plus Domain	Review	Setting	Population	Outcome of Interest	Policy Type	Intervention Details	Policy Effect	N° of relevant studies†
Plus 1: Age	(Stockings et al., 2016)	OECD HIC	Youth population	Drug use and prevention	Targeted	College education interventions, screening and brief interventions, motivational enhancement, peer self-help, family-based interventions, juvenile drug courts	*	291 (414)
					Universal	Sales restriction, police and law enforcement, scheduling under international conventions, mass-media, or education interventions	+/-	
Plus 2: Disabilities and co-morbidities	(Holmes et al 2019)	UK, USA	Individuals with mental health and drug use disorder	Drug use reduction	Targeted	Digital computer and mobile phone technology-assisted interventions for co-morbid depression and substance use	+	3 (6)
	(Dugdale et al., 2019)			Drug use reduction			+	33 (33)
	(Atun et al., 2011)					Integrated mental health and substance abuse services in community primary care setting	+	8 (12)
Plus 3: SUD	(Potier et al., 2014)	OECD HIC	Heavy drug users and PWID	Drug overdose	Universal	Provision of supervised injection services for safe drug use	+	75 (75)
	(Thomson et al., 2019)	USA		Injectable drug use risk - HIV		Syringe and needle exchange programs delivered through public health interventions in a community pharmacy setting	+	28 (15 reviews)
	(Fischer et al., 2015)	OECD HIC		Injectable drug use risk - HIV	Universal	Secondary material + environmental prevention programs	*	N/A
	(Bassuk et al., 2016)	USA	Drug users in recovery	Substance use	Targeted	Psychosocial treatment interventions	+	N/A
					Targeted	Targeted, community-based prevention interventions		N/A
					Targeted	Peer-delivered recovery support services for drug addictions	+	9 (9)
Plus 4: TDLE	(Janulis, 2012)		PWID	Injectable drug use risk - HIV	Universal	Pharmacy-delivered non-prescription syringe distribution services	+	29 (47)
	(Althaus et al., 2011)	All	Frequent ED users	Drug use frequency	Targeted	Clinical Case-management interventions for frequent ED users	+/*	3 (11)
	(Kouyoumdjian et al., 2015)		People in or released from prisoned	Drug use frequency		Motivational interviewing during imprisonment, educational and skills building programs during imprisonment, services after release	+	24 (95)
	(Ruff et al., 2010)	USA	Substance-abusing couples	Drug abuse and family violence		Behavioral Couples Therapy for substance abuse treatment	+	23 (23)

Figure 5. Table 4. Systematic reviews of interventions for individual PROGRESS-Plus factors. Effect key: + beneficial; – negative; +/- mixed; * inconclusive; 0 no effect. †Number of relevant studies to this umbrella review within a given systematic review/meta-analysis (total number of studies within a given review in parentheses).

Table 5: Systematic reviews of intervention effectiveness for multiple co-occurring PROGRESS-Plus factors

PROGRESS-Plus Domain	Review	Setting	Population	Outcome of Interest	Policy Type	Intervention Details	Policy Effect	N° of relevant studies†
Place of Residence + Plus 1: Age	(Hendricks Brown et al., 2007)	USA	Children in rural setting	Drug Use	Universal	School-based drug prevention programs	+	182 (182)
	(Coren et al., 2016)	USA, KOR	Street-connected children	Reduced drug use	Targeted	Interventions for promoting reintegration and harmful behavior reduction in street-connected youth	+/-	11 (13)
Place of residence + Plus 3: SUD	(Miler et al., 2020)	OECD HIC	Substance-abusing homeless individuals	Drug use frequency + reduction	Targeted	Peer-support provision to and from individuals with lived experience	+	34 (62)
	(Paquette & Pollini, 2018)	USA	PWID among non-urban dwellers	HIV risk attributable to injective drug use	Universal	Safe/supervised safe injection sites	+	26 (34)
					Targeted	Partner notification program for individuals who tested positive for HIV	+	
					Universal	Clean syringe access in non-urban pharmacies and other secondary sources, HIV testing and syringe exchange program HIV risk reduction interventions	-	
	(Hwang et al., 2005)		Homeless people with SUD	Drug use reduction	Targeted	Case management interventions, post-detoxification stabilization programs and therapeutic community	+/*	41 (73)
					Universal	Drop-in centres, community facilities	*	
Plus 1: Age + Education	(J. P. Davis et al., 2017)	OECD HIC	Emerging adults in non-college settings	Drug use reduction or prevention	Targeted	Prevention and treatment interventions, such as motivational interviewing, personalized feedback, and e-interventions	+	50 (50)
Plus 1: Age + Plus 3: SUD	(Carney & Myers, 2012)		Adolescents with SUD	Change in drug use	Targeted	Early, brief interventions with a drug use screening and intervention component	+	9 (9)
	(Li et al., 2016)			Drug use frequency and attitudes	Targeted	Motivational interviewing interventions in primary care setting, implemented through a person-centered, conversational style encouragement to reduce drug use	0	9 (10)
Plus 1: Age + Plus	(Salvo et al., 2012)		Adolescents with mental health and drug use disorders	Substance use prevention	Targeted	Skills-based primary prevention programs	+/-	8 (8)
					Universal	Secondary prevention interventions for substance use disorders in youth with mental disorders	+	
Gender/ Sexuality + Race/ Ethnicity/ Culture	(Valdez et al., 2018)	USA	Latino men	Drug use outcomes	Targeted	Gender adapted or culturally adapted interventions for SUD treatment compared with standard treatment	+/0	7 (12)
Gender/Sexuality + Plus 1: Age	(Coulter et al., 2019)		Sexual- + gender minority youth	Drug use reduction	Targeted	Community-based services with individualized treatment plans	+	2 (12)
PROGRESS-Plus Domain	Review	Setting	Population	Outcome of Interest	Policy Type	Intervention Details	Policy Effect	N° of relevant

								studies†
Gender/ Sexuality + Age + Race/Ethnicity/ Culture	(Hai, Hammock, et al., 2019)	USA, NLD	Women of childbearing age (18-45), by ethnicity	Prenatal and/ or postpartum drug use	Targeted	Dynamic technology-based interventions, delivered online or through text messages and apps, for preventing/decreasing drug use among women on childbearing age	*	5 (15)
Race/ Ethnicity/ Culture + Plus 1: Age	(Edwards et al. 2010) (Liddell & Burnette, 2017)	USA	Indigenous youth population	Drug use prevention, and refusal skills	Targeted	Culturally specific interventions for indigenous youth	*	4 (34)
					Universal	Culturally informed substance abuse interventions, such as those that avoid the separation of mind and body	+	14 (14)
					Universal	Multi-cultural interventions for drug use and misuse	-	
	(Snijder et al., 2020)	OECD HIC	Indigenous adolescents	Drug use prevention and use frequency	Targeted	Culture-based, culturally adapted programs for indigenous youth, such as language specific circle conversations, prayers and dancing in school, community and/or family setting	+	18 (26)
	(Mewton et al., 2018)		Non-white ethnicity youth population	Drug use prevention	Universal	Community-based and school-based programs involving peer-support training.	+	20 (52)
	(Lauricella et al., 2016)	USA			Targeted	Culturally grounded prevention interventions for ethnic minority youth	+	12 (33)
Race/Ethnicity/ Culture + Gender + Plus 1: Age	(O'Connor et al., 2020)	OECD HIC	Young people of different gender and ethnic profiles	Drug use prevention	Targeted	Culturally grounded hip-hop based interventions for African American youth	*	1 (23)
					Universal	Culturally grounded hip-hop based interventions for African American youth	+	
Place of residence + Plus 1: Age + Race/ Ethnicity/ Culture or Gender	(Wang et al., 2019)		Homeless youth of different gender and ethnic profiles	Drug use frequency	Targeted	Behavioral counselling interventions, (i.e., perinatal home visits to pregnant Native American youth, or girls in foster care)	+/-	29 (29)
					Universal	Universal drug prevention interventions (i.e., computer-based interventions)	+/*	
Race/ Ethnicity/ Culture + Plus 2: Disabilities and co-morbidities	(Guerrero et al., 2013)	USA	Latino population with mental health and drug disorders	Drug use frequency	Targeted	Skill building programs, individual and family level therapies	+	15 (22)
	(Leske et al., 2016)	OECD HIC	Indigenous adults with mental health and drug abuse disorders	Drug use frequency	Targeted	Structural interventions (such as housing support, drop-in centres, and shelters)	+/*	
					Universal	Culturally appropriate and language-adapted motivational enhancements and integrated treatments for dual disorders at community level	+	5 (100)
					Targeted	Culture-based and culturally adapted interventions for indigenous populations	*	7 (16)
					Universal	Interventions that have not been systematically modified for indigenous population context	*	
PROGRESS-	Review	Setting	Population	Outcome of	Policy	Intervention Details	Policy	N° of

Plus Domain				Interest	Type		Effect	relevant studies†
Plus 1: Age + Plus 4: TDLE	(Heerde et al. 2018)	USA	Youth leaving out-of-home care	Drug use frequency	Targeted	Transitional programs for youth leaving out-of-home care	0/*	4 (19)
	(Tolan et al., 2013)		Youth failing school or delinquent	Drug use reduction	Targeted	Mentoring interventions, where the mentee interacts with the professionally trained mentor over extended time	*	6 (46)
Plus 3: SUD + Plus 4: TDLE	(Sugarmen et al., 2020)		Incarcerated individuals with drug use disorders	Opioid use and overdose	Targeted	Interventions for prevention of opioid overdose fatality, such as screening brief interventions, referral to treatment and civil commitment	0	13 (13)
	(Brown, 2010)			Drug use reduction	Targeted	Drug treatment courts that target underlying drug addiction of criminal offenders through treatment service provision in conjunction with judicial supervision	+	5 (44)
Not peer reviewed	(Wittouck et al., 2013)	USA, AUS	Substance-using criminal offenders	Drug use	Targeted	Drug treatment courts and family treatment or dependency drug courts	+/*	10 (16)
	(Sabri et al., 2019)	All	Substance-abusing victims of	Drug misuse,	Targeted	Comprehensive, multi-component, peer-delivered and brief interventions with a combination of components	+	10 (17)

Place of residence

Eleven reviews assessed the effectiveness of illicit drug policy interventions for vulnerably housed or homeless individuals, inner-city and rural dwellers (see Tables 4 and 5). Of these, four reviews looked at place of residence as the sole domain of disadvantage, and seven stratified the subpopulation further, such as by concentrating on street-connected children (Coren et al., 2016), homeless individuals with existing substance use disorders (Miler et al. 2020, Hwang et al. 2005), or PWID and children in rural areas (Paquette & Pollini, 2018, Hendricks Brown et al. 2007).

Reviews that focused on homeless and street-connected individuals either evaluated the effectiveness of universal interventions in reducing drug use and/or related health comorbidities, such as through the provision of supervised injection sites, shelters, and clean syringe access in reducing illicit drug use frequency, or targeted interventions, such as peer support for people experiencing homelessness and case management interventions. Among these interventions, supervised injection sites showed a consistently positive effect across reviews on both drug use reduction and associated risk of HIV transmission for homeless and vulnerably housed individuals (Miler et al., 2020). Case management or reintegration interventions were either of mixed or inconclusive effectiveness for the homeless (Ponka et al. 2020, de Vet et al., 2013, Formosa et al. 2021, Coren et al., 2016). Peer support provision from people with past lived experience appeared effective at reducing drug use among the homeless (Miler et al., 2020).

Three reviews focused on intervention evaluation for rural dwellers. One looked at universal public health interventions in a community-setting and found no effect on drug use among individuals living in rural areas and inner-city areas. Another looked at the effectiveness of universal syringe exchange program, clean syringe access and HIV testing for PWID in non-urban pharmacies and found significant geographical barriers to access and worse illicit drug risk outcomes compared to urban dwellers (Paquette & Pollini, 2018). The same authors also found that partner notification programs for rural PWID who tested positive for HIV were effective in curbing HIV transmission. The last review for rural dwellers focused on universal school-based drug prevention programs for children in rural areas, and found that the interventions reduced drug use (Hendricks Brown et al., 2007).

Race/Ethnicity/Culture

Overall, 14 reviews, exclusively set either in USA, Canada or Australia, evaluated the effectiveness of interventions for individuals of different ethnic, racial, or cultural minorities, such as African American, Latino, or Indigenous subpopulations (see Tables 4 and 5). Five reviews focused on race, ethnicity, and culture as the sole disadvantage factor. The remaining 9 reviews attempted to combine ethnicity with additional co-occurring domains of disadvantage, such as gender (Valdez et al. 2018, Hai, Hammock, and Velasquez 2019), young age (Edwards et al., 2010, Mewton et al. 2018), or mental health and SUD co-morbidities (Guerrero et al., 2013, Leske et al.

2016).

Among the five reviews evaluating interventions for Indigenous subpopulations, most concentrated on targeted, culturally grounded, and deep structure prevention interventions that avoid the separation of mind and body, involving elements, such as language-adapted circle conversations, prayers and dancing in school, family and/or community setting (Snijder et al., 2020, Edwards et al., 2010, Liddell and Burnette 2017). The reviews found mostly positive, but some inconclusive evidence of effectiveness for culturally grounded interventions for Indigenous communities. Targeted behavioral counselling interventions, such as perinatal home visits to pregnant Native American youth, showed mixed program effectiveness (O'Connor et al., 2020). Only three reviews looked at how effective universal public health drug prevention interventions with no cultural adaptations are for Indigenous people. Two found inconclusive evidence of effectiveness (Leske et al. 2016, O'Connor et al., 2020), while the third reported an increase in drug use following a multicultural universal prevention intervention in Indigenous communities (Liddell & Burnette, 2017).

Latino and African American subpopulations appeared to have a similar, positive response to targeted interventions involving cultural and/or language adaptations (Lauricella et al., 2016). They were also especially effective for ethnic minority individuals with existing mental illnesses and SUD (Guerrero et al., 2013). Similarly, in terms of drug use prevention, ethnic youth appear to respond well to large-scale universal community- and school-based interventions that involve peer-support training (Mewton et al., 2018, O'Mara-Eves et al. 2015), but some reviews found inconclusive effectiveness of universal school-based interventions delivered through external consultants (Gorin et al., 2012). Both universal and targeted interventions aimed at preventing HIV and drug use among PWID, such as public health prevention programs and language-adapted referrals to social services in Hispanic communities showed evidence of effectiveness among racial, ethnic and cultural minorities (Herbst et al., 2007).

Occupation

Not considered in included studies.

Gender/Sexuality

Gender and/or sexual orientation was addressed as a potential co-occurring disadvantage factor in four reviews (see Tables 4 and 5). One looked at the effectiveness of gender- and culturally adapted interventions for Latino men, and found positive, albeit weak evidence of reduced drug use among the subpopulation compared to standard treatment (Valdez et al., 2018). Another looked at transgender and sexual minority youth and found two primary studies reporting positive results from community-based services with individualized treatment plans for sexual and gender minorities (Coulter et al., 2019). The third review looked specifically at effectiveness of interventions in reducing prenatal or postpar-

tum drug use among women of childbearing age and found inconclusive evidence of effectiveness of online technology-based interventions from five relevant primary studies (Hai, Hammock, et al., 2019). Lastly, while not explicitly focused on gender/sexual minority subpopulations, the fourth review looked at homeless youth as stratified by gender and found that homeless girls were likely to derive greater benefit from drop-in centres and shelters compared to boys in terms of reduced drug use frequency (Wang et al., 2019).

Education

One meta-analysis specifically focused on emerging adults in a non-college setting (J. P. Davis et al., 2017). The authors explicitly stated that most illicit drug interventions for emerging adults were evaluated within a college environment and individuals without university education may not only be at higher risk of harmful drug use but may also react differently to existing interventions than college students and graduates. The review authors found that targeted personalized feedback interventions, commonly associated with high effectiveness among emerging adults in college settings, had inconclusive results among their less educated peers with SUD. Other targeted interventions for individuals with SUD, such as motivational interviewing in community setting showed effectiveness among non-college educated emerging adults.

Socioeconomic Status

We identified one meta-analysis that explicitly examined community engagement drug intervention effectiveness for individuals of low socioeconomic status (O'Mara-Eves et al., 2015). This was part of a large-scale statistical synthesis of community-level public health interventions for a broad set of health behavioral outcomes among disadvantaged communities. The review reported that community engagement interventions targeted at people of low socioeconomic status were particularly effective at reducing existing illicit drug use.

Social Capital

Not measured in included studies.

Plus 1: Age

Twenty-two reviews examined differential effectiveness of illicit drug use interventions for at-risk age groups, such as youth population (see Tables 4 and 5). Of those, 6 looked at age without considering additional co-occurring PROGRESS-Plus domains and examined effectiveness specifically for children, adolescents, and young adults under 25 years old. The remaining 16 studies examined intervention with respect to at least one additional disadvantage factor, namely ethnicity, vulnerable place of residence (i.e., homelessness, rural setting), existing SUD and mental health problems, low educational attainment, sexual minority status and time-dependent life events (i.e., leaving out-of-home care or failing school).

Among reported intervention types, universal school-based educational interventions for prevention of illicit drug use were the most represented (Hale, Fitzgerald-Yau, and Viner 2014, Bonar et al. 2020, Hendricks Brown et al. 2007, Stockings et al. 2016). The effectiveness of these interventions appears to be inconclusive across studies, with multiple reviews coming to divergent conclusions regarding the effectiveness of these interventions. However, one large-scale review reported positive effects on drug use reduction among children in rural setting (Hendricks Brown et al., 2007). Generally, reviews reported that involving parents in the school community-based interventions through family-school partnerships appears to have a positive effect on drug use frequency among children and adolescents (Bonar et al. 2020, Hale et al. 2014). Other universal interventions included mass media prevention campaigns (O'Mara-Eves et al., 2015, Mewton et al., 2018) and public policy measures, such as police and law restriction (Stockings et al., 2016). These types of interventions showed either mixed or inconclusive findings. Targeted interventions reported in included reviews were generally brief, and included motivational interviewing for youth with SUD (Li et al., 2016), culturally adapted interventions for ethnic minority youth (Snijder et al., 2020), online interventions for women of childbearing age (Hai, Hammock, et al., 2019), or skill building and reintegration interventions for homeless or street-connected youth (Coren et al., 2016). Most of these targeted interventions showed inconclusive or null results, apart from culturally adapted interventions for Indigenous youth which are associated with decreased drug use among the subpopulations.

Plus 2: Disabilities and comorbidities

Disabilities and co-morbidities as a disadvantage domain category focused exclusively on the intersection of mental health and substance dependence in intervention evaluations. Overall, 7 reviews addressed mental health disorders, such as depression or suicidal thoughts for a range of subpopulations, including adolescents (Salvo et al., 2012), ethnic minorities (Guerrero et al., 2013, Leske et al., 2016), or victims of violence with post-traumatic stress disorders (Sabri et al., 2019).

One review reported that culturally adapted interventions for Latino males with dual disorders showed promising results of effectiveness in drug use reduction, compared to standard treatment (Guerrero et al., 2013). Another review reported inconclusive results of culturally adapted interventions for Indigenous adults with mental health problems (Leske et al., 2016). Comprehensive, multi-component interventions for substance-abusing victims of violence appeared effective at reducing severity of drug use (Sabri et al., 2019). Universal secondary prevention interventions appeared effective, while skills-based primary prevention programs demonstrated mixed effectiveness for adolescents with mental health difficulties and SUD (Salvo et al., 2012).

Other important disabilities that could contribute to increased illicit drug use and/or differential treatment effectiveness, including physical impairments, such as blindness, partial paralysis, or co-occurring serious illnesses such as cancers, car-

diovascular illnesses, or diabetes, were not addressed in any reviews.

Plus 3: Substance use disorders (SUD)

Thirteen reviews addressed individuals with SUD, such as existing alcohol or drug dependencies, in their intervention evaluations. Five of these looked at SUD as the sole PROGRESS-Plus domain irrespective of other characteristics, and eight included additional co-occurring risk-factors, such as young age (Li et al., 2016, Carney & Myers, 2012), vulnerable place of residence (Miler et al., 2020, Paquette & Pollini, 2018), or time dependent life events, including existing criminal charges (Brown, 2010).

Six reviews reported findings on the effectiveness of universal prevention interventions, such as safe injection sites or pharmacy-delivered needle-exchange programs, to reduce the risk of HIV transmission associated with injectable drug use and needle sharing. These programs appeared particularly effective for PWID in general and homeless individuals with existing SUDs in urban areas (Miler et al., 2020), but were not effective for PWID in rural areas due to a lack of geographical accessibility (Paquette & Pollini, 2018). Drug treatment courts generally appeared effective at reducing drug use among criminal offenders with SUDs (Brown, 2010, Wittouck et al., 2013).

Plus 4: Time-dependent life events (TDLE)

Time-dependent life events, such as recent release from prison, out-of-home facility, school failure or experience of domestic violence and substance use were addressed in nine reviews. All interventions reported in included reviews were targeted for the specific time-dependent life experience. Provision of drug treatment courts, social services and education in the periods leading up to and after release from prison appeared effective at reducing drug use frequency (Kouyoumdjian et al. 2015, Brown, 2010, Wittouck et al., 2013).

Similarly, behavioral couples therapy for co-abusing spouses and partners, as well as multi-component interventions for violence victims showed promising effectiveness in reducing drug use (Ruff et al. 2010, Sabri et al., 2019). Transition programs for youth leaving-out of home care did not appear effective at reducing drug use frequency (Heerde et al. 2018). No reviews looked at the effectiveness of universal interventions for individuals who recently experienced a significant and sudden life change, which could affect their current ability to access and benefit from the resources universally available.

4 Discussion

Overview of findings

This paper provided an overview of published systematic reviews reporting the effectiveness of illicit drug interventions for at least one PROGRESS-Plus subpopulation.

Over the last two decades, there has been a significant increase in the number of systematic reviews that evaluate effectiveness of illicit drug interventions with respect to different subpopulations at risk. However, we found that some subpopulations were not specifically considered in most, or all intervention evaluation systematic reviews within the field of illicit drug use, indicating that there is likely an insufficient production of primary studies evaluating interventions specifically for at-risk subpopulations. These subpopulations include individuals with low occupational and educational attainment, socioeconomic status, and social capital. We also found that within the broad PROGRESS-Plus domains represented in systematic reviews, some specific subpopulations are missing in intervention evaluation. For example, while age was a well-represented PROGRESS-Plus domain, no reviews looked at old age as a potential source of disadvantage both in terms of increased risk of drug use, and potentially differential intervention effectiveness, pointing to a limited number of primary studies evaluating drug use policy interventions for older at-risk population. Similarly, within the domain of Disabilities and Comorbidities, all reviews focused on mental health, and none addressed physical disabilities, such as blindness, deafness, or mobility impairments. Given the strong likelihood that drug use patterns differ markedly between persons with PROGRESS-Plus disadvantage traits relative to those without, this seems like a serious missed opportunity and a research gap for future researchers.

Moreover, we found that not all intervention types received the same attention when it comes to subpopulation-specific evaluation. Most reviews presented findings for interventions targeted at specific PROGRESS-Plus subpopulations, such as case management interventions for homeless individuals, or culturally adapted interventions for ethnic minorities, since these subgroups were most represented in primary studies of drug use intervention evaluations. Few reviews reported subpopulation-specific findings for large scale, universal interventions in the public health or policy domain, such as mass media campaigns or environmental prevention programs. Most reviews did not explicitly exclude universal interventions from their review focus, but there appears to be a lack of primary studies that examine universal illicit drug intervention effectiveness through the lens of PROGRESS-Plus subpopulations. This is an important finding, for it uncovers gaps in our current understanding of who benefits from existing universal interventions. For example, we have no systematic and synthesized understanding of how effective universal illicit drug interventions are for, among others, individuals who have been victims of violence, recently released from prison, or living with significant physical impairments. Understanding how universal interventions affect subpopulations is imperative for addressing persisting health inequities.

Apart from uncovering important gaps in subpopulation representation in knowledge syntheses within the field, we outlined general trends in the types of interventions that appear to be effective for specific subpopulations, as defined by their membership in one or more PROGRESS-Plus domains. Overall, we found consistent evidence of effectiveness of targeted, culturally-adapted and language-specific interventions for racial,

ethnic, and cultural minorities, who may not be able to benefit as much from standard universal interventions. Similarly, we found evidence from multiple reviews pointing towards effectiveness of drug treatment courts for criminal offenders with SUDs, and syringe exchange programs and safe injection sites for PWID.

While these reductions in drug use were consistent across reviews, most effects were reported as small to moderate and short-term. Long term reduction in harmful illicit drug use was seldom reported and when reported, the intervention effects were often not persistent over time. Reviews that reported mixed or inconclusive effectiveness of interventions suggest that rather than reducing illicit drug use, the interventions analysed may have shifted consumption patterns from hard drugs to softer drugs and alcohol (Coren et al., 2016).

Strengths and limitations

In undertaking this review, we faced challenges due to the heterogeneity of the included populations and interventions. For this reason, while we completed all necessary components of an umbrella review methodology, including a standardized data extraction form and a full quality appraisal using AMSTAR 2, our findings cannot offer a specific and unified set of effective intervention recommendation for illicit drug outcomes (Aromataris et al., 2020). This lack of concrete policy recommendations, commonly associated with focused umbrella reviews, was an anticipated limitation given our review design, since heterogeneity across all PICO elements has been identified as a major limitation of systematic reviews in social science (J. Davis et al., 2014). Instead, we produced an overview of subpopulation representation in systematic reviews within the field of illicit drug intervention evaluation, and general trends in the types of intervention design that appears to be effective for specific PROGRESS-Plus disadvantaged subgroups.

Another important challenge associated with heterogeneity of study designs included in this umbrella review, is that not all systematic reviews have a uniform risk of bias. Some reviews, for example, include primary study at higher risk of bias, such as cross-sectional studies or uncontrolled studies. The inclusion of such studies may have biased the intervention effectiveness results presented in the systematic reviews and subsequently in our umbrella review. However, the primary aim to provide an overview of existing reviews with subpopulation program evaluations remains unaffected.

While using PROGRESS-Plus proved to be an efficient way to cover a broad range of disadvantage characteristics and may represent a useful tool to standardize subpopulation intervention evaluation in future research syntheses, there are notable limitations associated with using the classificatory framework in its current form. Even though it offers researchers the flexibility to identify and add subject-specific disadvantage domains through the 'plus' component, some of the original PROGRESS domains may not work well for the purposes of subpopulation representation mapping analysis, because they do not offer mutually exclusive and separate domains. For example, disadvantaged occupational status, low educa-

tional attainment level and socioeconomic status are all separate domains within the framework, yet the two former domains could also be considered as dimensions and proxy measurements of the latter. This is not particularly troubling when using PROGRESS-Plus as a reference framework for potential sources of disadvantage, but it may be an important issue when using it as a classificatory framework for assessing subpopulation representation in intervention evaluations. This is because it may lead us to conclude that some disadvantage dimensions, such as socioeconomic status, may be underrepresented, while in fact the dimension representation has been dispersed across its common proxy measurements. In the context of our findings, however, this particular SES misclassification issue does not apply, for we can safely conclude that both SES and its common proxy measurements of occupational status and educational attainment are underrepresented in illicit drug intervention evaluation syntheses.

A further potential challenge to this review was publication bias. Prior to the review, and to inform the development of the 'plus' factors as part of the PROGRESS-Plus framework, we conducted an extensive literature review of existing evidence about disadvantage dimensions associated with excess risk of illicit drug use and associated comorbidities. However, there is a possibility that some potentially relevant subpopulations outside the PROGRESS-Plus framework have been left out of illicit drug research due to consistently insignificant findings and could not be identified during our initial literature review. Similarly, authors of primary studies of illicit drug intervention effectiveness may have failed to pick up statistically significant differences in effectiveness across subpopulations and have consequently chosen to exclude subpopulation analysis from the published findings. As a result, there may not be enough primary studies available to warrant the conduct of subpopulation-level intervention effectiveness systematic reviews. In parallel, any subgroup analysis, whether in primary studies or knowledge syntheses, should be based on plausible theory. It may be inappropriate or implausible for every study to estimate different effects within every PROGRESS-Plus domain. Sub-group analyses should also be planned a priori, and typically require large sample sizes. This may be challenging especially for small subgroups, such as sexual minorities. As a result, there are important cost implications associated with ensuring that sample sizes are sufficient to conduct relevant subpopulation analyses in primary studies, which may represent a pragmatic limit on the possibility of subpopulation analyses in systematic reviews and meta-analyses.

It is also important to note that most of the reviews included in this paper have not used a comparative methodology. Consequently, we cannot evaluate whether the policies deemed as effective for a particular subgroup are more, or less effective than other existing alternatives or whether the intervention effectiveness differs for the subpopulation at risk compared to a general population comparator. We cannot, therefore, make conclusions about the types of interventions most effective for specific subpopulations, but rather report general trends in effectiveness of interventions currently evaluated in existing systematic reviews. This inability to draw specific conclusions and tightly control for a comparative study design

of primary studies is a result of the necessary abstraction and loss of detail that resulted from our scoping focus on heterogeneous previous reviews, rather than primary studies. However, we found that such level of breadth and abstraction was necessary for our primary purpose of mapping the representation of subpopulations in illicit drug intervention evaluation systematic reviews. Ideally, we would have conducted a de novo review of all primary studies using a consistent methodology, but such a commitment was not possible given our time constraints.

Despite these limitations, the finding that some PROGRESS-Plus domains are not at all represented in existing systematic reviews of illicit drug interventions is both relevant and important to highlight not only the need to produce new impactful evidence syntheses of subpopulation-specific illicit drug program evaluations, but also to take a step back and ensure that enough primary research studies are being conducted to properly assess program effectiveness of drug use interventions across an array of different subpopulations at an elevated risk of drug use.

Implications for future research

Our findings shed light on existing gaps in the evaluation of illicit drug interventions for certain disadvantaged subpopulations. Future researchers in the field should examine the effectiveness of illicit drug interventions for individuals with vulnerable occupational status, social capital, lower educational attainment, and socioeconomic status in general. Moreover, most interventions evaluated in the included reviews were targeted, as opposed to universal. More research is needed to shed light on the effectiveness of universal policies on illicit drug outcomes among subpopulations at-risk, as contrasted with a general population comparator.

Moreover, researchers in other related policy evaluation fields should consider the potential existence of implicit biases and inequities from the failure to evaluate intervention effectiveness for all relevant stakeholders. We hope that the methodological approach used in this paper can be adapted to other policy evaluation areas and can present future equity-concerned researchers with an efficient and standardized way of evaluating whether their respective subfields assess existing interventions with respect to various relevant subpopulations at risk. Using PROGRESS-Plus has proved to be an effective way to quickly classify and map the extent of current subpopulation inclusivity in systematic reviews on illicit drug intervention effectiveness. The same approach can be adopted for other fields to potentially uncover broader evidence for persistent exclusion from impactful research across different health outcomes. While this review does not pertain directly to implementing evidence-based illicit drug interventions, it provides novel and relevant findings pointing to the need to evaluate interventions through the lens of an equity framework.

Conclusions

In conclusion, we found that systematic reviews of illicit drug intervention effectiveness were limited with respect to their focus on the following subpopulation characteristics: disad-

vantaged occupational status, low educational attainment, religion, socioeconomic status, and social capital, pointing to a potential gap in primary research production with respect to these subpopulations.

Among the most notable interventions evaluated in included reviews, community level, school- and family-based illicit drug intervention strategies appeared to be effective for youth populations. For racial, ethnic, and cultural minorities, we found a general trend in effectiveness of culturally relevant and language-adapted targeted interventions. Mass media and digital interventions have generally showed mixed effectiveness.

In their efforts to curb societal health inequities, academic researchers wishing to conduct future primary studies and systematic reviews alike in the field should consider including underrepresented subpopulations in upcoming intervention evaluations.

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Appendix A: PROGRESS-Plus factors and their relevance to illicit drug use

Place of residence. Residence in a disadvantaged area with potentially limited access to care (i.e., rural areas), high crime rates and drug availability (i.e., inner city urban areas), or homelessness may all negatively affect individuals' ability to resist drugs and seek help for misuse (Rhew et al. 2011, Mack 2017, Tyler et al. 2014).

Race/ethnicity/culture. Ethnic, racial, and cultural minorities or individuals who commonly speak a non-dominant language in their region of residence (i.e., individuals with African American, Hispanic, or Indigenous origins), may face discrimination and additional challenges related to illicit drug use and/or access to health and social services compared to their Caucasian counterparts (Newcomb et al. 2014, Delahunty and Putt 2006).

Occupation. Under-employment, unstable employment, work in unsafe conditions and joblessness, as well as a lack of employee benefits, insurance plans and other undesirable working conditions may negatively affect individuals' ability to resist drugs, seek help for substance use and recover (Henkel 2011, MacDonald and Pudney 2000).

Gender/sexual orientation. Gender self-identification and non-heterosexual orientation may affect individuals' perception of support available, exposure to violence, discrimination, drugs, and stigma associated with illicit drug use and help seeking (Newcomb et al., 2014).

Religion. Religiosity is commonly described as a protective factor against substance use in existing literature, affecting individual's perception of, and resilience to initiating illicit drug use (Dudley et al., 1987; Engs & Mullen, 1999). Due to the body of evidence suggesting this predominantly protective nature of religiosity when it comes to illicit drug use outcomes, we decided to omit religion as a disadvantage factor for drug use from our analyses.

Education. Individuals with low educational achievement level (i.e., high school degree or lower) may be less likely to benefit from existing interventions than their more educated peers. Low educational achievement is also a multifaceted disadvantage factor associated with a high probability of co-occurring occupational disadvantage, low socio-economic status, and residence in a disadvantaged area; all strongly associated with an increased risk of drug use and inability to access necessary help (Chatterji, 2006).

Socioeconomic status. Low socioeconomic status, commonly co-occurring with low education level, residence in a disadvantaged area, and occupational instability can affect the ability of individuals to access and/or afford health promoting resources and render them more vulnerable to harmful drug use than their wealthier counterparts (Gerra et al., 2020).

Social capital. Low social capital (i.e., insufficient community network, family relationships and loneliness) may affect the availability of social support and resilience to drugs. Insufficient social network may render individuals more vulnerable to peer pressure and may significantly affect successful recovery after drug use initiation (Åslund & Nilsson, 2013).

Plus 1: Age. Teenagers and young adults may be significantly more vulnerable to drug use initiation and peer pressure than other age groups, more likely to find access to and experiment with illicit drugs (Lynskey et al., 1999). Similarly, research evidence suggests that children and young persons may have less impulsivity control than older individuals, due to incomplete brain development (de Wit, 2009).

Plus 2: Disabilities and comorbidities. Disabilities, such as mental health disorders, chronic pain, diabetes, learning difficulties and physical handicaps among other things may significantly affect quality and self-perceived value of life. These comorbidities may contribute to increased risk of illicit drug use as a coping mechanism, while at the same time, drug use may lead to increased risk of developing further comorbidities, such as depression, chronic pain, or disabilities. They may also render individuals less likely to be able to access and benefit from available help (i.e., non-braille mass media interventions may be unsuitable for blind individuals) (Moore and Li 1998, Tyler et al. 2014).

Plus 3: Substance abuse disorders. Individuals with existing substance abuse disorders, such as alcohol, cigarette or soft-drug addictions may be more susceptible to initiate heavy drugs and develop addictions. They may also suffer from additional comorbidities associated with their existing substance use addition, such as depression (Carpenter et al., 2017).

Plus 4: Time-dependent life events (TDLE). Other at-risk subgroups include individuals during important life transitions, such as released and incarcerated criminal offenders, who may have history of drug abuse or violence and face societal stigma, discrimination, and inability to find suitable employment. They may likewise not possess the same ability to access resources than their counterparts without a criminal record. Additional groups facing disproportionate challenges are, for example, co-abusing partners and individuals living with or next to drug abusing relatives and roommates. These individuals have increased access to drugs and risk of peer pressure (Cavacuiti 2004, Mumola and Karberg 2006).

Appendix B: Search terms (excerpt)

Database: Ovid MEDLINE(R) ALL <1946 to March 31, 2021>

- 1 inequal\$.mp. or exp Healthcare Disparities/ or exp Health Status Disparities/ or exp Socioeconomic Factors/
- 2 Progress-Plus.mp. or exp PROGRESS PLUS/
- 3 (ethnic\$ or indigenous or non-white or Hispanic or black or African-American or asian).mp.
- 4 (socioeconomic status or employment status or unemploy\$ or social capital or occupation or religion).mp.
- 5 (place of residence or rural or urban or neighbourhood or homeless\$ or vulnerably housed or displaced or migrant or immigrant).mp.
- 6 (sexual orientation or LGBTQ+ or gay or lesbian or bisexual or transgender or queer or gender or sex).mp.
- 7 (at-risk age groups or youth or teenage\$ or elderly or old or pensioners).mp.
- 8 (disability or disabled or handicapped).mp.
- 9 (vulnerable group\$ or subpopulation\$ or at-risk population or addict\$ or comorbid\$ or mental health).mp.

10 (high-income or United States or Canada or EU).mp.
 11 (systematic review or scoping review or umbrella review or meta-analysis).mp.
 12 (public policy or policy or intervention\$ or health promotion\$ or program or campaign).mp.
 13 illicit drug.mp. or exp Illicit Drugs/ or cannabis.mp. or drug abuse.mp.
 14 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9
 15 10 and 11 and 12 and 13 and 14 (147 results)

The full search strategy (Embase, EPISTEMONIKOS, and Google Scholar queries) is available from the corresponding author upon request.

Appendix C: Studies excluded at full-text review, with reasons

Study	Reason for exclusion
(Ferri et al., 2013)	Findings duplicated in a review of reviews (Mewton et al., 2018)
(Fletcher et al., 2008)	Findings duplicated in a review of reviews (Mewton et al., 2018)
(Werb et al., 2011)	Findings duplicated in a review of reviews (Mewton et al., 2018)
(Sack & Thomasius, 2002)	Full text not in English
(Fernández et al., 2002)	Full text not in English
(Akanbi et al., 2020)	Intervention evaluation for a PROGRESS-Plus domain missing
(Hai, Franklin, et al., 2019)	Intervention evaluation for a PROGRESS-Plus domain missing
(King et al., 2015)	Intervention evaluation for a PROGRESS-Plus domain missing
(Kouyoumdjian et al., 2015)	Intervention evaluation for a PROGRESS-Plus domain missing
(Leece et al., 2019)	Intervention evaluation for a PROGRESS-Plus domain missing
(Mauri et al., 2020)	Intervention evaluation for a PROGRESS-Plus domain missing
(Werb et al., 2013)	Intervention evaluation for a PROGRESS-Plus domain missing
(Naar-King et al., 2009)	Wrong study design (not a systematic review)
(Rowan et al., 2015)	Wrong study design (not a systematic review)
(Russell et al., 2016)	Wrong intervention
(Sharma et al., 2020)	Full text not available
(Agabio et al., 2015)	Wrong outcome (illicit drugs not a separate outcome)
(Bagley et al., 2019)	Wrong outcome (effect on illicit drugs not quantified)
(Des Jarlais et al., 2013)	Wrong outcome (cannot link outcome to intervention)
(Lyles et al., 2007)	Wrong outcome (illicit drugs not a separate outcome)
(Mize et al., 2002)	Wrong outcome (illicit drugs not a separate outcome)
(Padmanathan et al., 2020)	Wrong outcome (cannot link outcome to intervention)
(Anderson et al. 2015)	Wrong outcome (insufficient evidence for illicit drug use)
(Bonell et al., 2013)	Wrong outcome (illicit drugs not a separate outcome)
(Carver et al., 2020)	Wrong outcome (effect on illicit drugs not quantified)
(Córdova et al., 2017)	Wrong outcome (insufficient evidence for illicit drug use)
(Getty et al., 2019)	Wrong outcome (medication adherence only)
(Stewart et al., 2018)	Full text not available
(Haldane et al., 2017)	Wrong outcome (illicit drugs not a separate outcome)

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